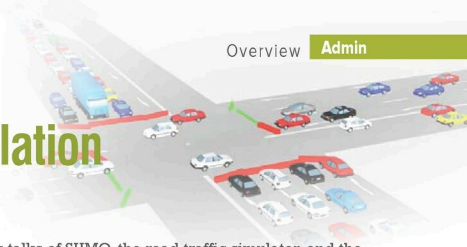

VANET Simulation Using FOSS

This second article in the series talks of SUMO, the road traffic simulator, and the software it needs to function well.

Deploying and testing VANETs involves high costs and a lot of time. Simulations of VANETs often require large and heterogeneous scenarios. The mobility behaviour of nodes in VANET significantly affects the simulation results. In a vehicular network, nodes (or vehicles) can only move along streets, prompting the need for a road model. Nodes in VANETs do not move independent of each other. They move according to well-established vehicular traffic models.

VANET mobility generators are used to generate traces of the vehicle's motion that can usually be saved and subsequently imported into a network simulator, in order to study the performance of the protocol or application. It is important to generate realistic movement traces in order to thoroughly evaluate VANET protocols, because in general, performance depends on the vehicles' movement traces. The inputs of the mobility generator include the road model, scenario parameters like maximum vehicular speed, vehicle arrival and departure rates, etc. The output of the trace contains the location of each vehicle at every instant for the entire simulation period, and their mobility profiles. Examples are SUMO, MOVE, FreeSim and VanetMobiSim.

Network simulators allow researchers to study how the network would behave under different circumstances. Users can then customise the simulator to fulfil their analysis needs. Network simulators are relatively fast and inexpensive, compared to the cost and time involved in setting up an entire experiment containing multiple networked computers, routers and data links. They allow researchers to test scenarios that might be difficult or costly to emulate with real hardware, especially in VANETs. Network simulators perform detailed packet-level simulation of source, destinations, data traffic transmission, reception, routes, links, and channels. These simulators are particularly useful to test new networking protocols, or to propose modifications to them. Examples are NS-2, GloMoSim and JiST/SWANS. Most existing network simulators are developed for MANETs, and hence require VANET extensions before they can be used to simulate vehicular networks.

Simulation is therefore the most common approach to develop or test new protocols for a VANET. Choosing the right simulation tools is an important step to get accurate predictions of real-world environments. In the previous article (published in the June 2012 issue of *LFY*), I had briefly discussed a road traffic simulator SUMO (Simulation of Urban MObility). For SUMO to function well, the following software are required:

1. Fedora 14 or Ubuntu 10.10
2. Java SDK 1.6
3. SUMO version 0.13.1 (for road traffic simulation)
4. Xerces version 3.0.1 (XML parser)
5. FOX tool-kit (GUI tool-kit)
6. PROJ (Cartographic Projection Library)
7. GDAL (Geospatial Data Abstraction Library)
8. NS-2 Version 2.34 (for actual network communication simulation)
9. MOVE

If you recall the previous article, the files needed to create traffic simulation mobility patterns using SUMO are:

1. *<Filename>.nod.xml* - Defines nodes (road junctions)
2. *<Filename>.edge.xml* - Defines edges (roads)
3. *<Filename>.netc.cfg*
4. *<Filename>.net.xml* - Generated using the net-convert command on the two files above.
5. *<Filename>.flow.xml* - Defines traffic flow definition
6. *<Filename>.rou.xml* - Defines vehicle routes
7. *<Filename>.sumo.cfg* - Simulation configuration

The SUMO command *net-convert* uses the node (1) and edge (2) definitions to generate the road network graph. In the flow files, you need to define the traffic flow characteristics, and the *.rou* file is used to define vehicle routes. All these files are mentioned in *sumo.cfg*, which is a configuration file for the generation of the final SUMO simulation. As seen in the above list, all the files are XML scripts. You need to write a tag for each junction, road segment or vehicle that is being defined for the simulation. For simulation of VANETs with a few vehicle nodes, that is fine—but in case a dense VANET is considered, generating all these files may become tedious and time-consuming. This is where MOVE comes into the picture.

MOVE (MObility generation for the Vehicular Environment) is intermediate software used between SUMO and NS2. It simplifies the generation of the XML files needed for SUMO simulation, and also converts the road traffic